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9. Operations on Regular Languages

1. UNION :- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
2. CONCATENATION : $-A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$
3. STAR : $A^* = \{x_1, x_2, x_3, \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$

Eg. $A = \{pq, r\}$, $B = \{t, uv\}$

$A \cup B = \{pq, r, t, uv\}$

$A \circ B = \{pqt, pquv, rt, ruv\}$

$A^* = \cancel{\{x_1 x_2 x_3 \dots}$

$\{\varepsilon, pq, r, pqr, rpq, pqpq, rr, pqpqpq, rrr, \dots\}$

Theorem1 : The class of Regular Languages is $\boxed{\text{closed}}$ under UNION

Theorem2 : The class of Regular Languages is $\boxed{\text{closed}}$ under CONCATENATION